Supplementary Material for "Adaptive Saturated ADRC of a Piezoelectric Drug-Delivery Actuator with Cubic Damping and Dahl Hysteresis"

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***Abstract*— This document collects the auxiliary proofs and the complete simulation study of the companion letter. Numbering of equations, assumptions, lemmas, propositions and the theorem follows that of the letter; symbols are as defined there.**

S-I. Proofs of the Auxiliary Lemmas

A. Proof of Lemma 1 (Dahl state bound)

Proof. At $z = \alpha/\beta$, $\dot{z} = -2\alpha x_2 \le 0$ for $x_2 \ge 0$ and $\dot{z} = 0$ for $x_2 < 0$; at $z = -\alpha/\beta$, $\dot{z} = 0$ for $x_2 \ge 0$ and $\dot{z} = 2\alpha|x_2| \ge 0$ for $x_2 < 0$. Hence $[-\alpha/\beta, \alpha/\beta]$ is positively invariant, and outside it the vector field points inward, so $|z(t)| \le \max\{|z(0)|, \alpha/\beta\}$ for all $t \ge 0$. □

B. Proof of Lemma 2 (Damping monotonicity)

Proof. $f_d(v) - f_d(w) = -\frac{c_p}{m_p}(v-w)(v^2+vw+w^2)$, and $v^2 + vw + w^2 = (v + \frac{w}{2})^2 + \frac{3}{4}w^2 \ge 0$. Hence $(f_d(v) - f_d(w))(v - w) = -\frac{c_p}{m_p}(v-w)^2(v^2+vw+w^2) \le 0$. Furthermore $v^2+vw+w^2-\frac{1}{4}(v-w)^2 = \frac{3}{4}(v+w)^2 \ge 0$, so $v^2+vw+w^2 \ge \frac{1}{4}(v-w)^2$, which yields (13). □

C. Proof of Lemma 3 (Derived disturbance-rate bound)

Proof. Since $\zeta = \xi$,

$$\dot{\zeta} = -\frac{k_p}{m_p}x_2 - \frac{\gamma}{m_p}\left[\sigma_\varepsilon'(x_2)\dot{x}_2 z + \sigma_\varepsilon(x_2)\dot{z}\right] - \frac{s_p}{m_p}\dot{F}.$$

By Proposition 1, $|x_2| \le \bar{x}_2$ and $|\xi| \le \bar{\zeta}_0$ by (26); by Lemma 1, $|z| \le \bar{z}$ and $|\dot{z}| = |-\alpha x_2 - \beta|x_2|z| \le \bar{z}_1$; by (14) and Assumption 1, $|\dot{x}_2| = |bu + f_d(x_2) + \xi| \le \bar{a}_2$; and $|\sigma_\varepsilon'| \le 1/\varepsilon_s$, $|\sigma_\varepsilon| \le 1$. Collecting the terms gives (27). □

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D. Proof of Lemma 4 (Local Lipschitz bound)

Proof. By construction, $\hat{x}_2^c \in [-\bar{x}_2, \bar{x}_2]$, and $x_2 \in [-\bar{x}_2, \bar{x}_2]$ by Proposition 1. On this interval, f_d is C^1 with $|f_d'(v)| = 3\frac{c_p}{m_p}v^2 \le L_d$. By the mean value theorem, $|f_d(\hat{x}_2^c) - f_d(x_2)| \le L_d|\hat{x}_2^c - x_2|$, and $|\hat{x}_2^c - x_2| \le |\hat{x}_2 - x_2|$ because the clamp is the projection onto $[-\bar{x}_2, \bar{x}_2]$ and x_2 belongs to this interval. Since x_2 lies in the clamping interval, $\hat{x}_2^c - x_2$ and $\hat{x}_2 - x_2$ have the same sign; as f_d is non-increasing, g has the opposite sign, so $g\tilde{x}_2 = -|g||\tilde{x}_2| \le -|g||d| = g\,d$, and (13) applied to $v = \hat{x}_2^c$, $w = x_2$ gives (29). □

E. Proof of Lemma 5 (Command and saturation-residual bound)

Proof. In (35), $\hat{b} \ge b_{\min} > 0$ and $|\ddot{y}_d| \le \bar{y}_2$. Since $\hat{\zeta} = \zeta + \tilde{\zeta}$ with $|\zeta| \le \bar{\zeta}_0$ (Proposition 1) and $|\tilde{\zeta}| = \omega_o^2|\varepsilon_3| \le \omega_o^2\bar{\varepsilon}$ (Proposition 2), $\hat{\zeta}$ is bounded. Likewise $|e_1| \le \bar{e}_1$ and

$$|\hat{e}_2| \le |e_2| + |\tilde{x}_2| \le \bar{x}_2 + \bar{y}_1 + \omega_o\bar{\varepsilon},$$

and $|\eta| \le \bar{\eta}$ by (34). The bound on Δu follows from the sector property $|\Delta u| \le |u_c|$ stated below (14). □

S-II. Proof of Proposition 1 (A priori boundedness of the plant)

Proof. The argument is a nested Lyapunov construction. Steps (i)–(ii) show that the forcing acting on the plant is bounded by a computable constant and build an energy function whose cross term saturates. Steps (iii)–(iv) bound x_1 and then x_2 in turn, and step (v) rules out finite escape. No property of the controller is invoked at any point.

(i) Bounded driving force. Write (1) as $m_p\dot{x}_2 = w(t) - c_p x_2^3 - k_p x_1$ with $w := m_p b\,u - \gamma\sigma_\varepsilon(x_2)z - s_p F$. By (14), $|u| \le u_{\max}$, and $|m_p b u| \le m_p b_{\max} u_{\max}$ by Assumption 1; by Lemma 1, $|z| \le \bar{z}$; and $|\sigma_\varepsilon| \le 1$, $|F| \le \bar{F}$. Hence $|w(t)| \le \bar{w}$ for all $t \ge 0$, *whatever the control law generating* u_c. The same bound holds verbatim for the original model (1b), since $|\mathrm{sgn}(x_2)| \le 1$ as well.

(ii) The Lyapunov function. Let

$$V_p := \tfrac{1}{2}m_p x_2^2 + \tfrac{1}{2}k_p x_1^2 + \kappa m_p \psi(x_1)x_2.$$

Minimizing V_p over x_2 gives

$$V_p \ge \tfrac{1}{2}x_1^2\left[k_p - m_p\kappa^2/(1+(x_1/x_s)^2)\right] \ge \tfrac{1}{2}(k_p - m_p\kappa^2)x_1^2 \ge 0$$

by the choice of κ. Since $|\psi| \le x_s$, Young's inequality gives $\kappa m_p|\psi||x_2| \le \frac{1}{4}m_p x_2^2 + c_0$, so

$$\tfrac{1}{4}m_p x_2^2 + \tfrac{1}{2}k_p x_1^2 - c_0 \le V_p \le \tfrac{3}{4}m_p x_2^2 + \tfrac{1}{2}k_p x_1^2 + c_0. \quad \text{(S1)}$$

(iii) Bound on x_1. Using $\psi' \in (0,1]$, $|\psi| \le x_s$ and $x_1\psi(x_1) = \rho(x_1) \ge 0$,

$$\begin{aligned}\dot V_p &= -c_p x_2^4 + w x_2 + \kappa m_p \psi' x_2^2 \\ &\quad + \kappa\psi w - \kappa c_p \psi x_2^3 - \kappa k_p \rho(x_1) \\ &\le -\Psi(|x_2|) + \kappa\bar w x_s - \kappa k_p \rho(x_1),\end{aligned} \quad \text{(S2)}$$

with Ψ as in (10). Therefore $\dot V_p < 0$ whenever $|x_2| > \bar s$ (then $\Psi(|x_2|) > \kappa\bar w x_s$ and $\rho \ge 0$) or $\rho(x_1) > (\Phi + \kappa\bar w x_s)/(\kappa k_p)$, i.e. whenever $|x_1| > X_1$; note that ρ is even and strictly increasing on $\mathbb{R}_{\ge 0}$, so ρ^{-1} is well defined. Hence $\dot V_p < 0$ outside the compact set $\mathcal{K} := \{|x_2| \le \bar s\} \cap \{|x_1| \le X_1\}$, on which both $V_p(0)$ and $\max_{\mathcal{K}} V_p$ are bounded by $\bar V_p$ by the right inequality in (S1). A trajectory can therefore never leave the sublevel set $\{V_p \le \bar V_p\}$, so $V_p(t) \le \bar V_p$ for all $t \ge 0$ and the first bound in (11) follows from the left inequality in (S1).

(iv) Bound on x_2. With $|x_1| \le \bar x_1$ established,

$$\begin{aligned}\frac{d}{dt}\big(\tfrac{1}{2}m_p x_2^2\big) &= -c_p x_2^4 + x_2\big(w - k_p x_1\big) \\ &\le |x_2|\big[(\bar w + k_p \bar x_1) - c_p|x_2|^3\big],\end{aligned}$$

which is negative whenever $|x_2| > \big((\bar w + k_p\bar x_1)/c_p\big)^{1/3}$. The comparison lemma then yields the second bound in (11).

(v) Global existence. The vector field of (3)–(5) is locally Lipschitz in the state and continuous in t, so a maximal solution exists; steps (iii)–(iv) bound it on its whole maximal interval of existence, which excludes finite escape. By the continuation theorem the solution is therefore defined for all $t \ge 0$.

(vi) Dependence on the plant data. Only a lower bound on c_p and upper bounds on m_p, k_p, γ and s_p are used in steps (i)–(iv): c_p appears in the dissipative term that is subtracted, while m_p, k_p, γ and s_p appear in $\bar w$ and in the coefficients of Ψ, all with a positive sign. Replacing the exact values by such bounds therefore preserves the conclusion and enlarges $\bar x_1$ and $\bar x_2$. □

S-III. Proof of Proposition 2 (Decoupled observer-error estimate)

Proof. Error dynamics. Subtracting (25) from (19),

$$\dot{\tilde X} = A_o\tilde X + E_1\big[g(\hat x_2^c, x_2) + \tilde b\,u\big] + E_o\dot\zeta, \quad \text{(S3)}$$

with $\tilde X = [\tilde x_1, \tilde x_2, \tilde\zeta]^\top$, A_o the observer-error matrix with characteristic polynomial (21), $E_1 = [0,1,0]^\top$, $E_o = [0,0,-1]^\top$ and $g := f_d(\hat x_2^c) - f_d(x_2)$; the $\hat b u$ term cancels exactly by (25), so no Δu term appears, while the gain mismatch survives as $\tilde b u$ because ζ does not contain it. Substituting $\ell_i = \omega_o^i\bar\ell_i$ and dividing the $\tilde x_2$- and $\tilde\zeta$-rows by ω_o and ω_o^2,

$$\dot\varepsilon = \omega_o\bar A\varepsilon + \frac{1}{\omega_o}\bar E_1\big[g + \tilde b\,u\big] - \frac{1}{\omega_o^2}\bar E_o\dot\zeta, \quad \text{(S4)}$$

so that, with $W := \varepsilon^\top\bar P\varepsilon$,

$$\dot W = -\omega_o\|\varepsilon\|^2 + \frac{2}{\omega_o}\varepsilon^\top\bar P\bar E_1\,g + \frac{2}{\omega_o}\varepsilon^\top\bar P\bar E_1\,\tilde b\,u - \frac{2}{\omega_o^2}\varepsilon^\top\bar P\bar E_o\dot\zeta. \quad \text{(S5)}$$

A structural identity. The columns of $\bar A$ are $\bar A e_2 = [1,0,0]^\top$ and $\bar A e_3 = [0,1,0]^\top$, so $(\bar P\bar A)_{:,2} = \bar P_{:,1}$ and $(\bar P\bar A)_{:,3} = \bar P_{:,2}$. Since $\bar A^\top\bar P + \bar P\bar A = (\bar P\bar A)^\top + \bar P\bar A$, its $(2,2)$ and $(3,3)$ entries are $2\bar P_{12}$ and $2\bar P_{32}$; equating them to -1 gives $\bar P_{12} = \bar P_{32} = -\frac{1}{2}$, irrespective of $\bar\ell$.

The damping mismatch. Splitting

$$\varepsilon^\top\bar P\bar E_1 = \bar P_{22}\varepsilon_2 - \tfrac{1}{2}(\varepsilon_1 + \varepsilon_3)$$

and using $\varepsilon_2 = \tilde x_2/\omega_o$,

$$\frac{2}{\omega_o}\varepsilon^\top\bar P\bar E_1 g = \underbrace{\frac{2\bar P_{22}}{\omega_o^2}\tilde x_2 g}_{\le -\frac{\bar P_{22}c_p}{2m_p\omega_o^2}d^4} - \frac{1}{\omega_o}(\varepsilon_1 + \varepsilon_3)g, \quad \text{(S6)}$$

where the first term is bounded through (29): this is where the monotonicity of the damping, rather than a Lipschitz estimate, is used. For the second term, $|\varepsilon_1 + \varepsilon_3| \le \sqrt{2}\|\varepsilon\|$ and $|g| \le L_d|d|$ by (28), so it is at most $A|d|$ with $A := \sqrt{2}L_d\|\varepsilon\|/\omega_o$. Writing $B := \bar P_{22}c_p/(2m_p\omega_o^2)$ and using $\max_{r\ge 0}(Ar - Br^4) = \frac{3}{4}A^{4/3}(4B)^{-1/3}$, the two terms of (S6) are together bounded by $a\|\varepsilon\|^{4/3}$ with $a = \frac{3}{4}(\sqrt{2}L_d/\omega_o)^{4/3}(4B)^{-1/3}$; finally

$$\max_{r\ge 0}\big(ar^{4/3} - \tfrac{\omega_o}{2}r^2\big) = \tfrac{16a^3}{27\omega_o^2} = C_g,$$

which absorbs the whole nonlinear contribution into the constant (30) at the price of half of the quadratic margin.

The remaining inputs. By Cauchy–Schwarz together with $|\tilde b| \le \bar b$ and $|u| \le u_{\max}$, the gain-mismatch term in (S5) is at most $(\bar c_b/\omega_o)\|\varepsilon\|$, and the $\dot\zeta$ term is at most $(\bar c_o/\omega_o^2)\|\varepsilon\||\dot\zeta|$. With $|\dot\zeta| \le \bar\zeta_1$ from Lemma 3,

$$\dot W \le -\frac{\omega_o}{2}\|\varepsilon\|^2 + K_o\|\varepsilon\| + C_g. \quad \text{(S7)}$$

Conclusion. Young's inequality gives

$$K_o\|\varepsilon\| \le \frac{\omega_o}{4}\|\varepsilon\|^2 + K_o^2/\omega_o,$$

so

$$\dot W \le -\frac{\omega_o}{4}\|\varepsilon\|^2 + D_\varepsilon \le -\frac{\omega_o}{4\lambda_{\max}(\bar P)}W + D_\varepsilon.$$

The comparison lemma yields

$$W(t) \le \max\{\lambda_{\max}(\bar P)\varepsilon_{0,\max}^2, 4\lambda_{\max}(\bar P)D_\varepsilon/\omega_o\}$$

and $\limsup_t W \le 4\lambda_{\max}(\bar P)D_\varepsilon/\omega_o$; (32) follows from

$$\lambda_{\min}(\bar P)\|\varepsilon\|^2 \le W \le \lambda_{\max}(\bar P)\|\varepsilon\|^2.$$

□

S-IV. Simulation Results

The closed-loop system (3) is simulated in Python. The eight-state closed loop $(x_1, x_2, z, \hat{x}_1, \hat{x}_2, \hat{\zeta}, \eta, \hat{b})$ is integrated with the implicit Runge–Kutta solver `Radau` of `scipy.integrate.solve_ivp`, with relative and absolute tolerances 10^{-9} and 10^{-12} and a maximum step size of 2×10^{-4} s. Since the nominal error is of the order of the absolute tolerance, the nominal run was repeated with tolerances 10^{-11} and 10^{-15}; the reported error is unchanged to five significant digits, so it is not limited by the integrator. The internal variables z, h, $F(t)$, and the true gain b are used only for plant simulation.

The desired displacement is $y_d(t) = 30\sin\left(\frac{2\pi}{10}t\right)$ μm, with analytical derivatives, and the unknown environmental force is $F(t) = 2\times10^{-3}e^{-0.15(t-8)}S(t-8)+10^{-3}\sin(0.7t)S(t-15)$, where $S(\cdot)$ is a smooth approximation of the Heaviside step function. Parameters are listed in Table S1. The envelope bounds $\bar{x}_1, \bar{x}_2$ used in the clamp (22) are evaluated from Proposition 1. The analytical constants $\bar{\zeta}_1$ and $\bar{u}_c$ are global worst cases rather than predictions of the observed errors. Six scenarios are reported: nominal tracking; a 58 μm reference that drives the actuator into sustained saturation; the same saturated scenario with a fixed and an adapted $\hat{b}$; a sweep of the observer bandwidth; measurement noise and plant-parameter mismatch; and a comparison with two baselines.

The adaptive estimate $\hat{b}(t)$ can be initialized at any $\hat{b}(0) \in [b_{\min}, b_{\max}]$: the midpoint $\hat{b}(0) = \frac{1}{2}(b_{\min} + b_{\max})$ is a simple choice that requires no experiment, whereas a more accurate, data-driven value is obtained from a short-pulse calibration test. Applying the voltage inputs $u_e \pm \delta u$ from the same operating condition, the stiffness, hysteresis, viscous, and environmental terms act as common-mode contributions, so that $x^+(t) - x^-(t) = b\,\delta u\,t^2 + \mathcal{O}(t^3)$ for $t \in (0, T]$ with T sufficiently small, and

$$\hat{b}(0) = \frac{\sum_{i=1}^{N} t_i^2\,[x^+(t_i) - x^-(t_i)]}{\delta u \sum_{i=1}^{N} t_i^4}, \qquad t_i \in (0, T], \tag{S8}$$

using only the applied voltage and the measured displacement. The cubic damping is not common-mode, so the pulse must be short enough that $\frac{c_p}{m_p}(2b\,\delta u\,T)^3 \ll b\,\delta u$. Here we take $\hat{b}(0) = 0.2333$, updated online by (15) inside $[b_{\min}, b_{\max}]$, which contains the true value $b = 0.2381$ with comfortable margin. The sensitivity to this choice is asymmetric, since $\hat{b}$ divides the command in (35): an oversized $\hat{b}$ weakens the loop and degrades tracking roughly in proportion, whereas undersizing $\hat{b}$ is considerably less detrimental. With $\hat{b} = 0.05$ the post-transient error is 1.8×10^{-6} μm at the same peak voltage and with no increase in noise sensitivity, while $\hat{b} = 2$ gives 5.6×10^{-2} μm. When b is poorly known, a conservative choice at the lower end of the interval is therefore preferable.

All states are initialized at zero except $\hat{b}(0) = 0.2333$; in particular $x_1(0) = x_2(0) = z(0) = 0$ for the plant and $\hat{x}_1(0) = \hat{x}_2(0) = \hat{\zeta}(0) = \eta(0) = 0$ for the compensator.

Nominal tracking. Fig. S1 shows accurate tracking of the sinusoidal reference, with a position error below 9.0×10^{-6} μm. Fig. S2 shows that $\hat{\zeta}$ follows $\zeta = \xi$; the applied voltage stays within ±30.5 V, well inside $\pm u_{\max}$, and the closed-loop input-output map remains close to the ideal line $y = y_d$. The velocity estimate stays below 19 μm/s, well inside $\bar{x}_2$, so the clamp (22) is never active, and $|\eta| \le 1.2 \times 10^{-11}$, eight orders of magnitude below the a priori bound (34), so the leakage has a negligible numerical effect in these runs. As the third panel of Fig. S2 shows, $\hat{b}(t)$ remains in $[b_{\min}, b_{\max}]$

TABLE S1: Plant, adaptation, control, and observer parameters used in simulation. Controller and observer gains follow the bandwidth parameterization $(s+\omega_c)^3$ and $(s+\omega_o)^3$, for which $k_I = \omega_c^3 < 9\omega_c^3 = k_1 k_2$ holds by construction. The adaptation gain is large because the regressor $\tau = 2u(P_c\hat{\chi})_2$ is expressed in SI units, where the tracking errors are of order 10^{-8} m.

Parameter	Value	Parameter	Value
m_p	8.4×10^{-3} kg	c_p	5.0×10^{10} N s^3/m^3
d_p	2.0×10^{-3} N/V	k_p	2.0×10^{3} N/m
γ	1.0×10^{-6}	α	1.1773×10^{6}
β	121.9874	s_p	1
$b = d_p/m_p$	0.2381	$\hat{b}(0)$	0.2333
$b_{\min}$	0.15	$b_{\max}$	0.35
γ_b	3×10^{7}	$u_{\max}$	60 V
$\bar{x}_1$	189 μm	$\bar{x}_2$	225 μm/s
λ_I	0.1 s^{-1}	ε_s	5×10^{-6} m/s
ω_c	4.0×10^{3} rad/s	ω_o	1.2×10^{4} rad/s
$k_1 = 3\omega_c^2$	4.8×10^{7}	$k_2 = 3\omega_c$	1.2×10^{4}
$k_I = \omega_c^3$	6.4×10^{10}	$\ell_1 = 3\omega_o$	3.6×10^{4}
$\ell_2 = 3\omega_o^2$	4.32×10^{8}	$\ell_3 = \omega_o^3$	1.728×10^{12}

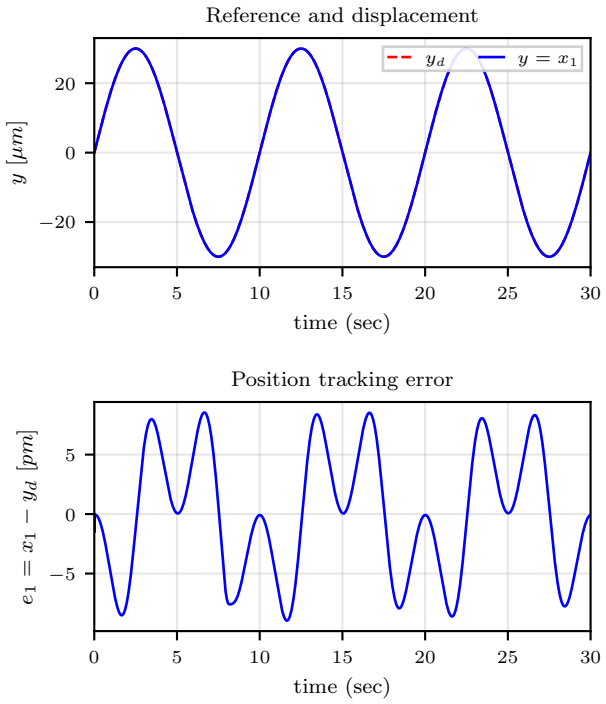


Fig. S1: Closed-loop tracking response under the adaptive saturated ADRC with the gains of Table S1. Top: reference y_d and actuator displacement $y = x_1$. Bottom: position tracking error $e_1 = x_1 - y_d$, in picometres (1 pm $= 10^{-6}$ μm).

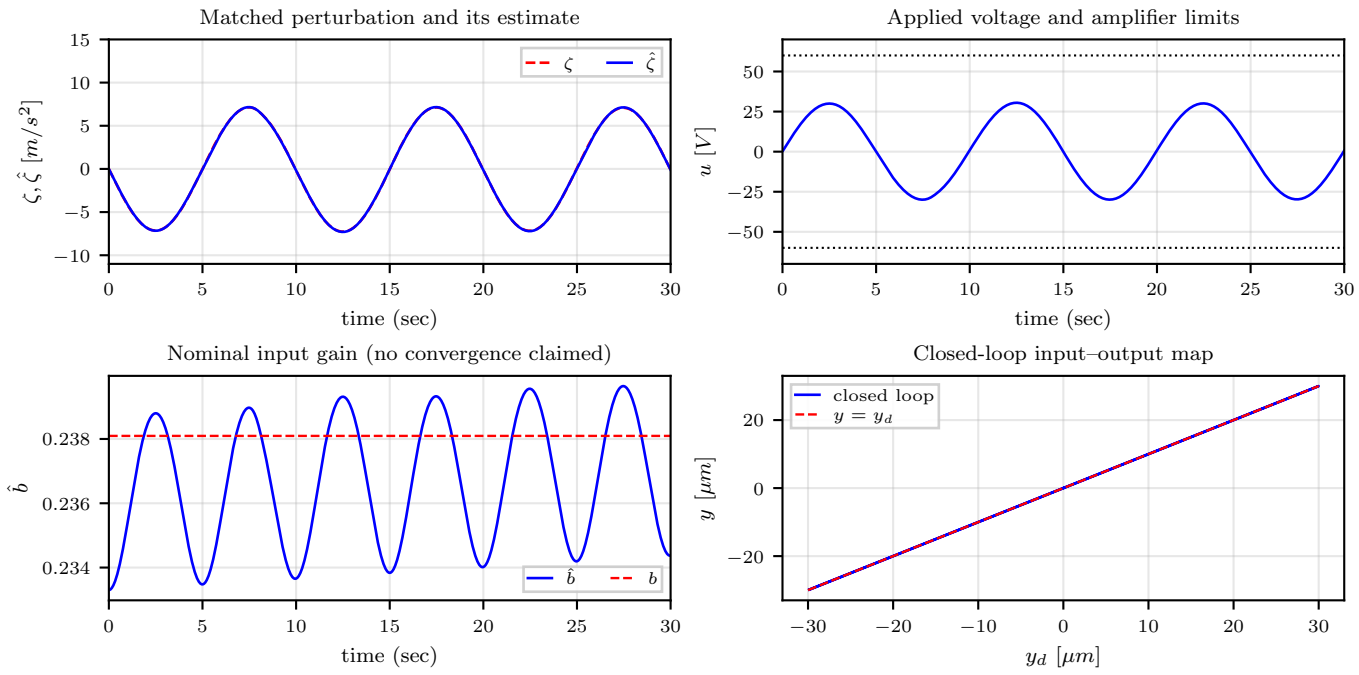


Fig. S2: Nominal run, $y_d = 30\sin(2\pi t/10)$ μm. Top left: the ESO estimate $\hat{\zeta}$ tracks the matched plant perturbation $\zeta = \xi$, which collects the stiffness, hysteresis and environmental-force terms. Top right: the applied voltage stays within ± 30.5 V, so the amplifier limits $\pm u_{\max}$ are never reached and the saturation is inactive in this run. Bottom left: the nominal input gain $\hat{b}$ oscillates with the reference and remains close to its initial value, slightly below the true gain b (dashed); no convergence is claimed. Bottom right: the closed-loop map (y_d, y) is indistinguishable from the ideal line $y = y_d$ at this scale.

and remains close to its initial value: the ESO absorbs the input-gain mismatch into $\hat{\zeta}$, so the tracking error carries no information on $\tilde{b}$ and the regressor is negligible. Convergence of $\hat{b}$ to b is neither claimed nor observed, in agreement with Theorem 1.

Active saturation. Fig. S3 repeats the experiment with a 58 μm reference, close to the static stroke $d_p u_{\max}/k_p = 60$ μm. The unsaturated command reaches 133 V, and the actuator is saturated during 42% of the run, where the saturation fraction is the fraction of the horizon for which $|u_c| > u_{\max}$. This definition is used because sat_s is a smooth compression, so Δu is not identically zero outside that region. Although $|\Delta u|$ reaches 74.8 V, the position error stays below 7.1×10^{-3} μm, where throughout this document the reported error is $\max|e_1|$ over $t > 5$ s and this run uses the calibrated $\hat{b}(0)$ of Table S1 with the adaptation enabled. The error concentrates in bursts at the *positive* peaks of the reference, although the saturation windows are symmetric in depth and duration. The asymmetry comes from the Dahl term: under $(x_1, x_2, z) \mapsto (-x_1, -x_2, -z)$ the state z is odd but the force $h = \gamma\,\mathrm{sgn}(x_2)z$ is even, so h keeps its sign whereas every other term of (1) changes sign. Consequently, the Dahl term introduces a bias that opposes positive motion; repeating the run with $\gamma = 0$ makes the two peaks equal.

Fixed versus adapted input gain. Fig. S4 compares, in the saturated scenario, a fixed $\hat{b} = b_{\max}$ with the adaptive law started from the same value. With $\hat{b}(0) = b_{\max}$ the post-transient error drops from 1.48×10^{-2} μm for the fixed-gain case to 1.00×10^{-2} μm with adaptation, as $\hat{b}$ moves from 0.350 toward 0.337; the adapted run also improves upon the fixed calibrated-gain case, whose error is 1.14×10^{-2} μm. Under deep saturation the time-varying adaptation therefore provides a greater benefit than starting from the best fixed value. The estimate does not converge to b, and the improvement is not explained by the mean value of $\hat{b}(t)$ either. What the adaptive law does is respond during the saturation events, flattening the error peaks that dominate the post-transient measure. In the unsaturated scenario the two are indistinguishable, because the ESO then absorbs the input-gain mismatch through $\hat{\zeta}$ and the tracking error carries almost no information on $\tilde{b}$. The adaptation is thus effective precisely when saturation prevents the observer from absorbing the mismatch on its own.

Observer bandwidth. Fig. S5 sweeps ω_o from 10^3 to 4×10^4 rad/s. The post-transient position error falls from 5.8×10^{-3} μm at $\omega_o = 10^3$ to 8.7×10^{-6} μm at 4×10^4, most of the gain being obtained by $\omega_o = 1.2 \times 10^4$, the value of Table S1. Even the slowest observer keeps the error at 0.019% of the reference amplitude, more than two orders of magnitude below the dosing tolerance of the application, so reducing ω_o by more than a decade costs accuracy but does not compromise the task. Over the range considered here, the perturbation-estimation error is limited primarily by the input-gain mismatch rather than by the bandwidth: the extended state converges to $\hat{\zeta} \to \zeta - \tilde{b}u$, so $|\tilde{\zeta}| \to |\tilde{b}||u|$, which is why $|\tilde{\zeta}|$ drops by two orders of magnitude once the adaptation has reduced $|\tilde{b}|$. The same mechanism appears in the derived bound through the term $\bar{c}_b/\omega_o$ of K_o, which dominates $\bar{c}_o\bar{\zeta}_1/\omega_o^2$ only beyond $\omega_o \approx 9 \times 10^6$ rad/s, so below that crossover the bound is conservative

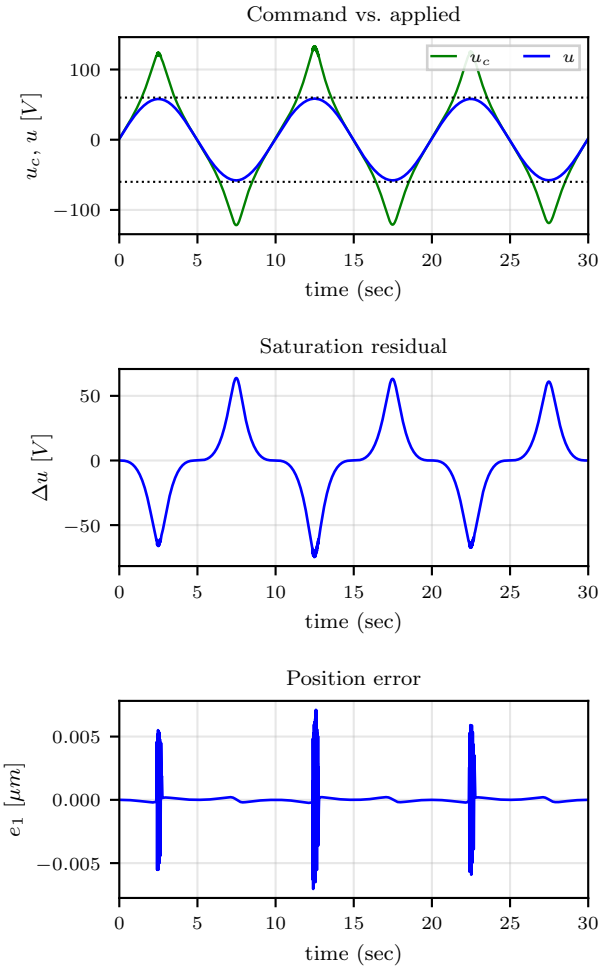


Fig. S3: Active saturation (58 μm reference): unsaturated command u_c and applied voltage u, saturation residual Δu, and position error, from top to bottom. The actuator is saturated during 42% of the run.

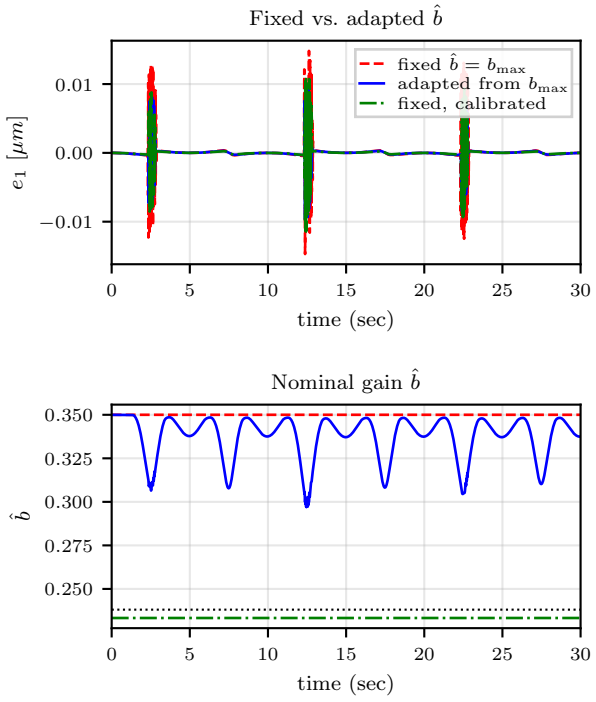


Fig. S4: Fixed versus adapted nominal input gain under active saturation. Top: position error for $\hat{b}$ held fixed at $b_{\max}$, for the adaptive law started from the same value, and for the calibrated fixed value. Bottom: the corresponding $\hat{b}(t)$, with the true gain b dotted. In the unsaturated scenario the curves are indistinguishable.

and does not capture this effect.

Robustness. Fig. S6 reports measurement noise and plant-parameter mismatch. With band-limited sensor noise of $\sigma =$ 1 nm and 10 nm the error is 4.3×10^{-3} and 5.2×10^{-2} μm, i.e. of the order of the injected noise, as expected for a high-bandwidth position loop. A $+20\%$ mismatch in c_p and k_p gives 1.8×10^{-5} μm and a -20% mismatch gives 4.2×10^{-6} μm, with the voltage between 24.5 and 36.5 V. Varying the cubic damping over a $50\times$ range and the hysteresis coefficient γ by $20\times$ changes the error by less than 3%, since both terms are absorbed by ξ and estimated by the ESO.

Baselines. Fig. S7 compares the proposed scheme with the ADRC of Flores and Rakotondrabe, ref. [12] of the letter, designed for linear viscous damping, and with a PID with filtered derivative that uses the position measurement only and no observer. All three are tuned for the same closed-loop bandwidth of 4×10^3 rad/s and draw the same peak voltage; the linear damping coefficient supplied to ref. [12] is the describing-function equivalent $\frac{3}{4} c_p (A\omega)^2 = 13.3$ N s/m of the nominal orbit, which is the most favourable choice for it. At the nominal input gain the post-transient errors are 9.0×10^{-6}, 8.5×10^{-2} and 2.0×10^{-2} μm. When the true d_p is 30% lower than its nominal value the proposed scheme degrades to 3.6×10^{-5} μm, whereas ref. [12], which cancels k_p, c_p and d_p explicitly, degrades to 1.14 μm. Both baselines improve if their gains are raised, which is why the bandwidth is matched.

From displacement to delivered dose. The displacement x of the second gap in Fig. 1 acts on a chamber of effective area A_e, so the delivered flow, volume and dose error are

$$Q(t) = A_e\, x_2(t), \qquad V = A_e\, \Delta x, \qquad \delta V = A_e\, e_1, \quad \text{(S9)}$$

and the *relative* dose error $\delta V/V = e_1/\Delta x$ is independent of A_e. For a chamber of 1 mm diameter ($A_e = 7.9 \times 10^{-7}$ m^2) the nominal run delivers 47.1 nL per stroke at a peak flow of 0.89 μL/min, which is the operating range reported for piezoelectric micropumps in drug delivery, ref. [15] of the letter. The corresponding dose errors are 7.1×10^{-6} nL for the proposed scheme, 1.6×10^{-2} nL for the PID and 6.7×10^{-2} nL for ref. [12]; with the input gain 30% off, the latter grows to 0.89 nL. In relative terms these are 1.5×10^{-5}%, 3.3×10^{-2}%, 1.4×10^{-1}% and 1.9% of the stroke volume: all remain below the 5% dosing tolerance reported for micropumps in ref. [15], but with margins spanning five orders of magnitude.

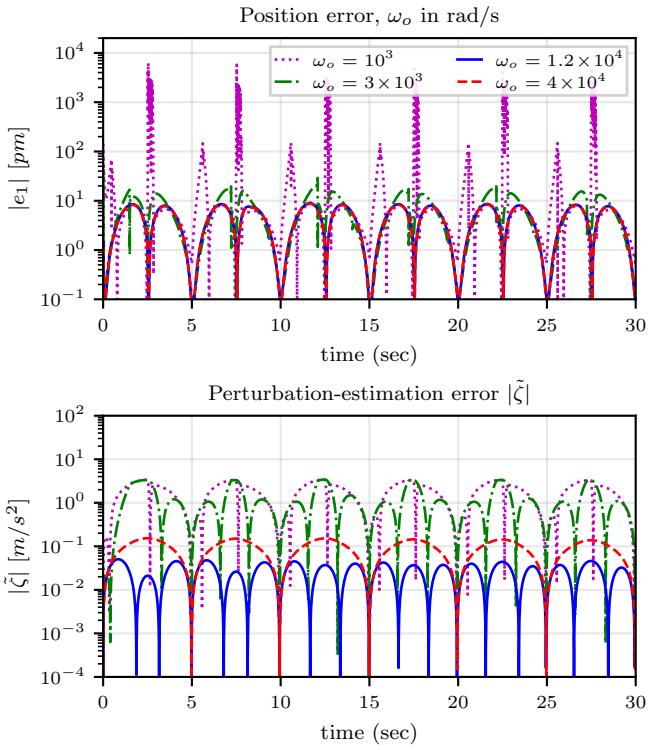


Fig. S5: Effect of the observer bandwidth; both panels use a logarithmic scale, since the slowest case exceeds the fastest by more than three decades. No lower bound on ω_o is imposed by Theorem 1. Top: position error in picometres (1 pm = 10^{-6} μm). Bottom: perturbation-estimation error.

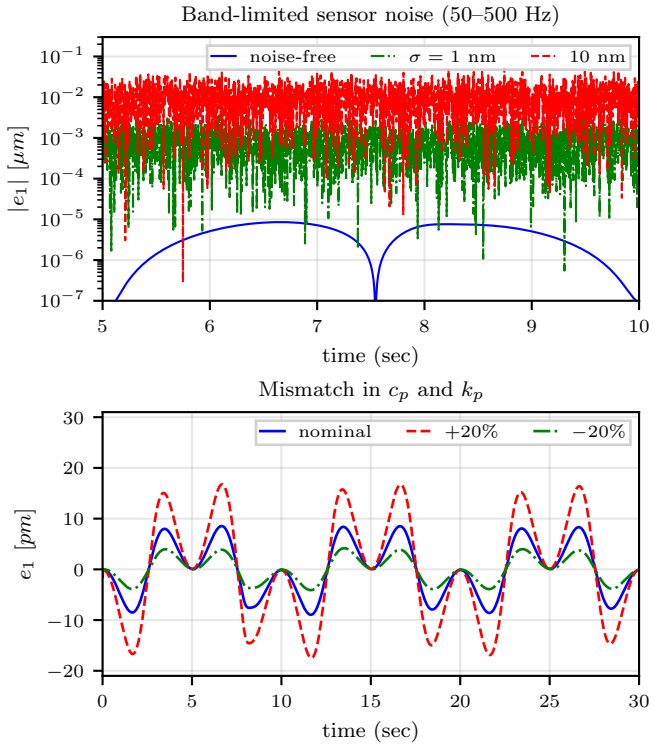


Fig. S6: Robustness. Top: band-limited (50–500 Hz) sensor noise, 10 s horizon. Bottom: $\pm 20\%$ mismatch in c_p and k_p, 30 s horizon. Same solver and tolerances as the remaining runs.

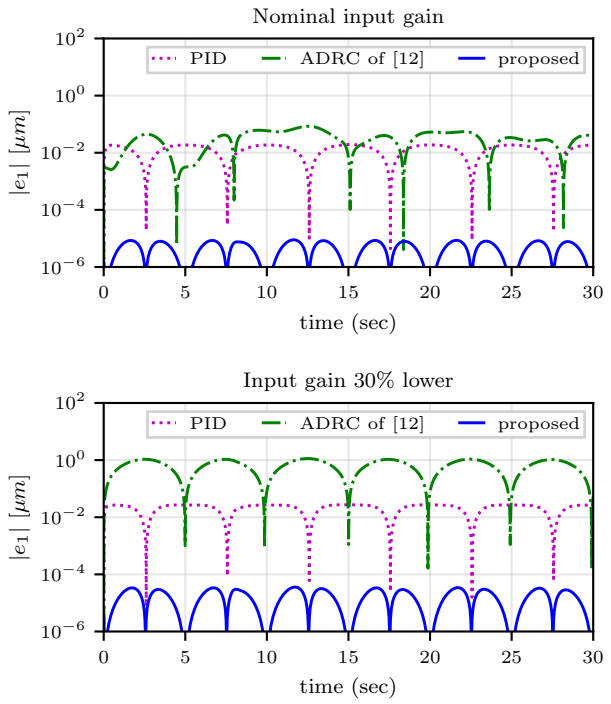


Fig. S7: Comparison with the ADRC of Flores and Rakotondrabe, ref. [12] of the letter, and with a PID with filtered derivative using the position measurement only (no observer). All three are tuned for the same closed-loop bandwidth and draw the same peak voltage. Top: nominal input gain. Bottom: the true d_p is 30% below its nominal value.